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COMPUTER

CHAPTER 26

Microsoft Excel

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CHAPTER 26

Microsoft Excel

Computer Awareness

IN THIS CHAPTER

- Workbooks and worksheets
- Cells and references
- Ribbon tools
- Formulas and functions
- Data tools
- Errors and shortcuts

•MACRO = RECORD TASKS

- Default Extension = .docx

Microsoft Excel (MS Excel) is a spreadsheet application developed by Microsoft. It is used to store, organize, calculate, analyze, and visualize data in the form of rows and columns.

MS Excel is one of the most widely used data processing and data analysis tools in offices, businesses, schools, banks, and government organizations.

MS Excel is a spreadsheet software used to perform calculations, create tables, analyze data, generate charts, and prepare reports.

Nature of MS Excel = Spreadsheet Application / Data Processing Software

WHAT IS A SPREADSHEET?

A spreadsheet is a table made up of rows and columns.

The intersection of a row and a column forms a cell.

SPREADSHEET STRUCTURE

Cell = Intersection of Row and Column

Example:

Cell Address = B3

(B = Column, 3 = Row)

WORKBOOK, WORKSHEET AND CELL

Workbook

An Excel file is called a Workbook.

Example

Book1.xlsx

Worksheet

A single sheet inside a workbook is called a Worksheet.

Example

Sheet1, Sheet2, Sheet3

Cell

A rectangular box formed by the intersection of a row and a column.

Example

A1, B5, D10

RELATIONSHIP

Workbook

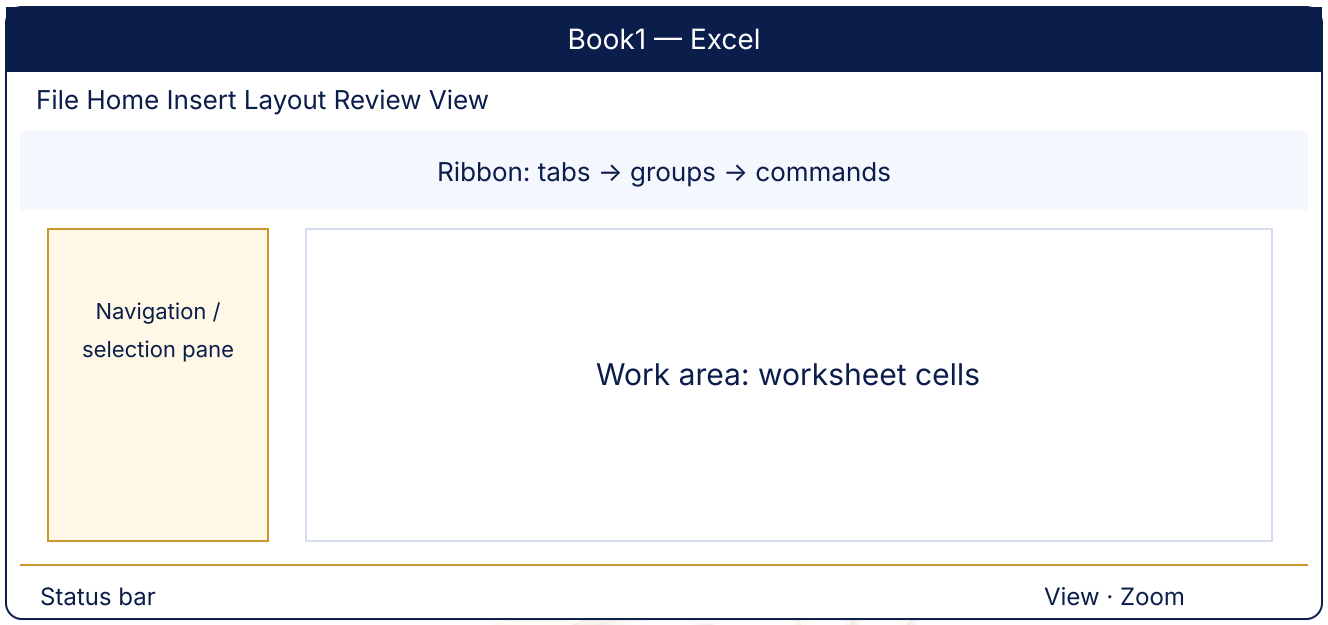
- Worksheet 1
- Worksheet 2
- Worksheet 3
- Cells

Exam Point

Collection of Cells → Worksheet

Collection of Worksheets → Workbook

STRUCTURE OF EXCEL WINDOW



Schematic window structure; exact controls vary by version.

IMPORTANT PARTS OF EXCEL

1. Name Box

Displays the address of the currently selected cell.

Example If cell D5 is selected:

Name Box = D5

2. Formula Bar

Used to enter and edit formulas and functions.

Example

=SUM(A1:A10)

3. Column Heading

Columns are identified by letters.

A B C D E F ...

Last Column

XFD

Total Columns

16,384

Exam Point

Excel Columns = A to XFD

4. Row Heading

Rows are identified by numbers.

1

2

3

4 Last Row

1,048,576

Exam Point

Total Rows = 1,048,576

5. Sheet Tabs

Displays all worksheets in the workbook.

Example:

- Sheet1
- Sheet2
- Sheet3

6. Spreadsheet Area

The working area where data is entered.

7. Zoom Control

Used to zoom in and zoom out.

Range

10% to 400%

RIBBON TABS IN EXCEL

MS EXCEL

- Home
- Insert
- Page Layout
- Formulas
- Data

- Review
- View

HOME TAB

The Home Tab contains the most frequently used commands.

Groups in Home Tab

Home

- Clipboard
- Font
- Alignment
- Number
- Styles
- Cells
- Editing

1. CLIPBOARD GROUP

Contains:

- Cut
- Copy
- Paste
- Format Painter

Format Painter

Copies formatting from one cell to another.

2. FONT GROUP

Used for formatting text.

Options

- Font Style
- Font Size
- Bold
- Italic
- Underline
- Font Color

- Fill Color

3. ALIGNMENT GROUP

Controls alignment of cell contents.

Features

- Left Align
- Center Align
- Right Align
- Top Align
- Bottom Align

Wrap Text

Displays long text on multiple lines within a cell.

Example:

Gur

preet

Singh

Merge & Center

Combines multiple cells into one cell and centers the data.

Example:

A1+B1+C1

Merged Cell

4. NUMBER GROUP

Used to format numbers.

Examples

- Currency (₹)
- Dollar ($)
- Percentage (%)
- Date
- Time

5. STYLES GROUP

Contains:

Conditional Formatting :Formats cells according to conditions.

Example:

Marks > 90 → Green

Marks < 40 → Red

Cell Styles :Predefined formatting styles.

Format as Table :Converts selected data into a table.

6. CELLS GROUP

Used for:

- Insert Cells
- Delete Cells
- Format Cells

7. EDITING GROUP

Contains:

Sort :Arranges data.

Example:

A to Z

Z to A

Filter :Displays only selected data.

Find & Replace :Search and replace data.

Clear :Removes contents or formatting.

INSERT TAB

Used to insert various objects.

1.Tables

Creates tables.

Pivot Table :A special table used to summarize, analyze, and organize large amounts of data.

Example Sales data of 1000 employees summarized into one report.

Exam Point

Pivot Table = Summary of Complex Data

2.Filters

Slicer :Visual filtering tool for Pivot Tables.

Timeline :Filters Pivot Table data based on dates.

3.Charts

Creates:

- Bar Charts
- Pie Charts
- Line Charts
- Area Charts

4.Text Group

Contains:

- Header
- Footer
- WordArt
- Signature Line

NoteMost of the features in the Insert tab are similar to the Insert tab in MS Word, so you can learn them from there.

PAGE LAYOUT TAB

Controls page appearance.

The Page Layout tab in Microsoft Excel allows users to control how their workbook will appear when printed or viewed in different layouts. The key sections are:

1. Themes

- Changes the overall appearance of the workbook.
- Includes fonts, colours, and effects that give a consistent look across

all sheets.

2. Page Setup

- Margins: Adjusts the white space around the page.
- Orientation: Choose between Portrait (default) or Landscape.
- Page Breaks: Splits the worksheet into multiple printable pages.

3. Sheet Options

- :

◦: Toggle the display of worksheet gridlines. ◦: Show or hide row and column headings.

4. Arrange

- The Arrange group on the Page Layout tab arranges worksheet objects, such as shapes and pictures.

Workbook-window arrangement belongs to the View tab's Window group.

These settings help in controlling the printing and presentation of data effectively.

FORMULAS TAB

Contains all formulas and functions.

FUNCTION

A predefined formula in Excel is called a Function.

Example =SUM(A1:A10)

Exam Point

Function = Predefined Formula

Common Functions

1.SUM

Adds numbers.

=SUM(A1:A10)

2.AVERAGE

Finds average.

=AVERAGE(A1:A10)

3.MAX

Largest value.

=MAX(A1:A10)

4.MIN

Smallest value.

=MIN(A1:A10)

5.COUNT

Counts numerical entries.

=COUNT(A1:A10)

Exam Point :Every Formula Starts With "="

Example:

=A1+B1

DATA TAB

Used for data management.

Features

- Sort
- Filter
- Data Validation
- Remove Duplicates
- External Data Import

The Data tab in Microsoft Excel is one of the most important sections for managing, analysing, and organising data efficiently. It provides a wide range of tools to import, sort, filter, and process data from various sources. Here is an overview of the key groups and commands in the Data tab:

1. Get & Transform Data (Get External Data)

- Get Data: Allows you to import data from external sources like databases, web pages,

text/CSV files, XML, JSON, or other workbooks.

- Connections: Manage the links between your Excel file and connected data sources.
- Refresh All: Update data from connected sources.

2. Sort & Filter

- Sort Ascending / Descending: Arrange data either from smallest to largest or largest to

smallest.

- Custom Sort: Sort data by multiple columns or custom order.
- Filter: Add filters to columns to view only the data you need.
- Clear: Remove applied filters.
- Reapply: Reapply the last used filter settings.

3. Data Tools

- Text to Columns: Split data in a single column into multiple columns based on

delimiters (like commas or spaces).

- Remove Duplicates: Quickly remove repeated values.
- Data Validation: Restrict the type of data that can be entered in a cell (like dates,

numbers, or specific lists).

- Consolidate: Combine data from multiple ranges.
- What-If Analysis: Includes tools like Scenario Manager and Goal Seek to analyse

different outcomes.

4. Forecast (Data Analysis)

- Forecast Sheet: Create projections and future trends based on historical data.

5. Outline

- Group / Ungroup: Arrange data in collapsible sections.
- Subtotal: Automatically calculate sums, averages, or other aggregations for grouped

data. 6. Queries & Connections (Power Query)

- Lets you manage all data queries in one place, especially for large or complex datasets.

The Data tab is essential when working with large datasets, performing analysis, and keeping your Excel workbook organised and dynamic.

REVIEW TAB

Similar to MS Word Review Tab.

Features

- Spelling Check
- Comments
- Track Changes
- Protect Sheet
- Protect Workbook

1. Proofing

- Spelling: Checks the worksheet for spelling errors.
- Thesaurus: Offers synonyms for selected words.
- Research/Smart Lookup: Provides definitions, context, and related information for the

selected text.

2. Accessibility

- Check Accessibility: Scans the workbook for elements that may be difficult for people

with disabilities to read or use.

3. Comments

- New Comment: Inserts a comment in a cell to provide notes or instructions.
- Show/Hide Comment: Displays or hides comments in selected cells.
- Show All Comments: Displays all comments in the worksheet.
- Delete Comment: Removes a comment from a cell.
- Previous/Next Comment: Navigates through the comments in the worksheet.

4. Notes (In newer versions of Excel)

- New Note: Adds a traditional note to a cell.
- Edit Note: Modifies existing notes.
- Convert Notes: Converts old-style notes to modern comments.

5. Protect

- Protect Sheet: Locks the worksheet to prevent users from changing data or structure.
- Protect Workbook: Secures the workbook structure to prevent adding, deleting, or

moving sheets.

- Allow Edit Ranges: Specifies which cells can be edited while the sheet is protected.

The Review tab is mainly for data verification, collaboration, and security. For team workbooks, this tab helps monitor changes, leave feedback, and ensure the document is error free before sharing.

VIEW TAB

Workbook Views

- Normal View: Default View
- Page Break Preview :Shows page breaks.
- Page Layout View :Shows worksheet as printed pages.
- Custom View:Creates customized worksheet views.

Show Group

Show/Hide:

- Formula Bar
- Gridlines
- Headings

Window Group

Freeze Panes :Locks rows and columns while scrolling.

Example

Keep first row visible while scrolling.

Exam Point

Freeze Panes = Lock Rows and Columns

CELL ADDRESS

A cell address is identified by: Column + Row

Example

D5

D = Column

5 = Row

RANGE OF CELLS

A group of cells is called a Range.

Example

A1:C3

Meaning :Includes all cells from A1 to C3. Exam Point :Range is represented using Colon (:)

DEFAULT SETTINGS

Feature	Value
Default Workbook Name	Book1
Default Orientation	Portrait
Default View	Normal
Default File Extension	.xlsx
Mouse Pointer Shape	Plus (+)

DATA ALIGNMENT

Data Type	Default Alignment
Text	Left
Numbers	Right

Exam Point

Text → Left

Numbers → Right

EXCEL FILE EXTENSIONS

Extension	Purpose
.xls	Older Excel Workbook
.xlsx	Default Workbook
.xlsm	Macro-enabled Workbook
.xla	Add-in
.xlam	Macro-enabled Add-in
.xlw	Workspace File
.csv	Comma Separated Values
.ods	OpenDocument Spreadsheet
.txt	Text File
.html	Web Page

COMMON EXCEL ERRORS

Error	Meaning
#DIV/0!	Division by Zero
#NAME?	Unknown Function Name
#VALUE!	Wrong Data Type
#REF!	Invalid Cell Reference
#NUM!	Invalid Number
#N/A	Value Not Available
#####	Column too narrow, or a negative date/time value

Exam Point

Most Common Error = #DIV/0!

WORKBOOK VS WORKSHEET

Workbook	Worksheet
Excel File	Sheet Inside Workbook
Example: Book1.xlsx	Example: Sheet1

SORT VS FILTER

Sort	Filter
Rearranges Data	Hides Unwanted Data
A→Z	Show Specific Records

Shortcut keys	Description
Ctrl + Right Arrow	Move to right edge of current data region
Ctrl + Left Arrow	Move to left edge of current data region
Ctrl + Up Arrow	Move to upper edge of current data region
Ctrl + Down Arrow	Move to lower edge of current data region
shift+arrows	To select cells
Ctrl + N	Create a new workbook
Shift + F11	Create new sheet
Ctrl +1	Format cell dialog box
Ctrl +shift + +	To add new column and row
Ctrl + -	To delete selected row and column

Shortcut keys	Description
alt+page up/down	Horizontal scrolling
Page up/down	Vertical scrolling
ctrl+page up/down	To shifts between sheets
Ctrl +shift +page up/down	Select the current and adjacent worksheet
Ctrl +space	To select entire column
Shift + space	To select entire row
Ctrl +shift+space	Select current region or entire worksheet, depending on context
Ctrl +home	To move to first cell
Ctrl +end	Move to last used cell
Ctrl +shift +f	Open Font tab of Format Cells

Shortcut keys	Description
Ctrl + Shift + End	Select from active cell to last used cell
Ctrl + G	Open Goto dialog box
Ctrl +F11	To add macro sheet
Alt + enter	To write in next line in same cell
Enter	Move cursor to next cell
F11	To add chart in different sheet
Ctrl +shift +U	To expand formula bar
Ctrl + tab ,ctrl +f6	To switch between workbooks
Ctrl +shift +$	To add currency sign
ctrl +shift+#	To show date format

Shortcut keys	Description
Ctrl +shift +@	To show time format
ctrl + ;	Current date
Ctrl +shift+:	Current time
Ctrl + t	To create table
Ctrl +enter	Fill selected cells with the current entry
F1	HELP
Ctrl + f1	To hide ribbon
ctrl +shift + f1	Hide all bar
F2	To edit active cell
Shift + f2	Insert or edit a note (legacy comment)

Shortcut keys	Description
Ctrl +shift +%	To add % sign
Ctrl + 1	Cell dialog box
Shift + f3	Insert function dialog box
Shift +f5	To open find and replace box
ctrl +5	Strike through
F7	Check spelling
ctrl + 9	Hide row
Ctrl +shift+9	Show row
F11	To insert chart
Alt +F11	Open the Visual Basic Editor; Alt + F1 inserts an embedded chart

Shortcut keys	Description
Shift +F11	To add new sheet

Formulas	Description
Sum(A1:E1)	It add all the number which include A1,b1,c1,d1,e1
sum(A1,E1)	It add the numbers which include A1 ,E1 only
Min()	To find minimum number
Max()	Maximum number
average()	Average of numbers
if(condition,print1,print2)	If condition satisfied print1 otherwise print 2
Rank()	To find rank from selected cells
CONCATENATE(A2,B2)	Join the two cell values into one text string
now()	Today date and time
today()	Present date only

Formulas	Description
count()	No of cells which have numerical value
countA()	It count no of cells which have numerical, alphabet value
countblank()	Count number of blank cells
FREQUENCY(A1:C1,bins_array)	Return counts in numerical bins; not simply duplicate counts
lookup()	Used for look ups. Value from a one row and one column
VLookup()	Look up a value in the leftmost column of a table
Hlookup()	Look up a value in the top row of a table
Sum product	First multiply then add
clean()	It avoids unprintable character
trim()	Remove unwanted spaces in text

Formulas	Description
lower()	Convert the character of text in lower case
Upper()	Convert to upper case
PROPER(text)	Capitalize the first letter of each word
len()	It return the length of a text
find()	Find a character in the text
REPT(text,number_times)	Repeat text the specified number of times
SUBSTITUTE / REPLACE	SUBSTITUTE replaces matching text; REPLACE replaces characters by position
EXACT(text1,text2)	Case-sensitive comparison of two text strings
T(value)	Return text if the value is text; otherwise return an empty string
Sqrt()	Square root

Formulas	Description
LARGE(array,k)	Return the kth-largest numerical value
SMALL(array,k)	Return the kth-smallest numerical value
round()	Round of the numbers who are in points

MOST IMPORTANT EXAM FACTS

1.MS Excel is a Spreadsheet Application.

2.Default workbook name = Book1.

3.Default extension = .xlsx.

4.Total columns = 16,384 (A–XFD).

5.Total rows = 1,048,576.

6.Cell = Intersection of Row and Column.

7.Formula starts with "=".

8.Function = Predefined Formula.

9.SUM is the most commonly used function. 10.Pivot Table summarizes large data.

11.Freeze Panes locks rows and columns.

12.Merge & Center combines cells.

13.Conditional Formatting formats data automatically.

14.Text aligns left by default.

15.Numbers align right by default.

QUICK REVISION NOTES

- MS Excel = Spreadsheet Software
- Workbook = Excel File
- Worksheet = Sheet
- Cell = Row + Column Intersection
- Default File = Book1
- Default Extension = .xlsx
- Columns = A to XFD
- Rows = 1,048,576
- Formula starts with "="
- SUM = Addition Function
- Pivot Table = Data Summary
- Sort = Arrange Data
- Filter = Display Specific Data
- Conditional Formatting = Format Based on Conditions
- Freeze Panes = Lock Rows/Columns
- Merge & Center = Combine Cells
- Text Left, Numbers Right

MICROSOFT POWERPOINT (MS POWERPOINT)

Microsoft PowerPoint (MS PowerPoint or PPT) is a presentation graphics software developed by Microsoft. It is used to create professional presentations in the form of slides.

PowerPoint helps users present information visually using:

- Text
- Images
- Audio

- Video
- Charts
- Animations
- Transitions

MS PowerPoint is a presentation software used to create, edit, organize, and display information through electronic slide shows.

Exam Point

Nature of MS PowerPoint = Presentation Graphics Software